TASK 3: CUSTOMER DEMOGRAPHIC ANALYSIS

Python • Pandas • Matplotlib • Detailed guide with sample data and code

1. Objective

Analyse tour enquiry data to identify popular destinations and explore patterns by customer age and income bracket. Exploratory data analysis (EDA) describes patterns in the available data; it does not by itself explain why customers choose a destination.

2. Questions to answer

- Which destinations receive the most enquiries?
- What is the age distribution of customers who enquire?
- How do enquiry counts vary across age groups and income brackets?
- Are there duplicate, missing, inconsistent, or implausible records?

3. Suggested columns and synthetic sample

One row should represent one tour enquiry. Adapt the names to match your supplied CSV.

enquiry_id	age	annual income (INR)	destination	enquiry date	status
1001	24	450000	Goa	2026-01-08	New
1002	36	900000	Kerala	2026-01-09	Converted
1003	29	620000	Goa	2026-01-10	New

These sample rows are fictional. Keep income in one currency and on one time basis, such as annual INR.

4. Cleaning workflow

- Inspect row/column counts, data types, duplicates, and missing values before changing anything.
- Trim whitespace and standardize destination labels (for example, 'goa' becomes 'Goa').
- Convert age and income to numeric values and parse dates as dates.
- Investigate impossible ages and negative income; document any corrections or exclusions.
- Remove exact duplicates only when they are genuinely duplicate enquiries.

```
import pandas as pd

df = pd.read_csv("tour_enquiries.csv")
df.columns = df.columns.str.strip().str.lower()
df = df.drop_duplicates()
df["destination"] = df["destination"].astype("string").str.strip().str.title()
df["customer_age"] = pd.to_numeric(df["customer_age"], errors="coerce")
df["annual_income"] = pd.to_numeric(df["annual_income"], errors="coerce")
df["enquiry_date"] = pd.to_datetime(df["enquiry_date"], errors="coerce")

df.loc[~df["customer_age"].between(1, 100), "customer_age"] = pd.NA
df.loc[df["annual_income"] < 0, "annual_income"] = pd.NA
print(df.shape)
print(df.isna().sum())
print(df.describe(include="all"))
```

5. Age groups and income brackets

Bands make comparisons easier. These thresholds are examples; choose appropriate bands for your data and document them.

```
df["age_group"] = pd.cut(
    df["customer_age"], bins=[17, 25, 35, 45, 100],
    labels=["18–25", "26–35", "36–45", "46+"]
)
df["income_bracket"] = pd.cut(
    df["annual_income"], bins=[-1, 500000, 1000000, float("inf")],
    labels=["< ■5L", "■5–10L", "> ■10L"]
)

print(df["destination"].value_counts())
print(pd.crosstab(df["destination"], df["age_group"]))
print(pd.crosstab(df["destination"], df["income_bracket"]))
```

6. Charts

- Bar chart: enquiry count by destination, sorted from highest to lowest.
- Histogram: customer age distribution.
- Optional stacked bar chart: destination by age group or income bracket.
- Label axes clearly and save charts at 200 dpi or higher.

```
import matplotlib.pyplot as plt

df["destination"].value_counts().sort_values().plot(kind="barh")
plt.title("Tour Enquiries by Destination")
plt.xlabel("Number of enquiries")
plt.ylabel("Destination")
plt.tight_layout()
plt.savefig("destination_popularity.png", dpi=200)
plt.show()

df["customer_age"].dropna().plot(kind="hist", bins=10, edgecolor="black")
plt.title("Customer Age Distribution")
plt.xlabel("Age")
plt.ylabel("Number of enquiries")
plt.tight_layout()
plt.savefig("customer_age_histogram.png", dpi=200)
plt.show()
```

7. Reporting your findings

Example wording only: “Goa has the largest number of enquiries in this sample. The age histogram shows the largest cluster between 26 and 35 years.” Replace this with what your actual output shows.

- Enquiry counts are not the same as completed bookings.
- Missing income values can affect demographic comparisons.
- Report the number of records analysed and any exclusions.
- Do not claim causation from descriptive charts alone.

Task 3 deliverables

- Cleaned CSV file.
- Python notebook or .py script.
- Destination bar chart and age histogram.
- Age/income crosstabs, written findings, and limitations.
- README with column definitions and steps to reproduce the analysis.